IPV4 subnet calculator using Powershell

Objective

To develop a PowerShell-based IPv4 subnet calculator capable of computing the subnet mask, network address, broadcast address, first host, last host, and total usable hosts from an IPv4 address and CIDR notation.

Introduction

Subnetting divides an IP network into smaller logical networks. This experiment demonstrates subnet calculations using Windows PowerShell and bitwise networking concepts.

Tools Required

- Windows 10/11
- Windows PowerShell
- Visual Studio Code or Notepad

Execution Policy Issue

Initially, PowerShell blocked script execution with the message 'running scripts is disabled on this system'. This was resolved by opening PowerShell as Administrator and running:

Set-ExecutionPolicy -Scope CurrentUser RemoteSigned

After confirming with 'Y', the script executed successfully.

Procedure

Step 1: Create the Project Folder

A project directory named Subnet-Calculator was created on the Desktop to store the PowerShell script and all related project files.

Step 2: Create the PowerShell Script

A PowerShell script named SubnetCalculator.ps1 was created inside the project folder. The script accepts an IPv4 address and CIDR prefix as input and calculates subnetting information automatically.

Step 3: Open Windows PowerShell

Windows PowerShell was opened and navigated to the project directory using:

-*cd C:\Users\computer1\Desktop\Subnet-Calculator*

```
PS C:\WINDOWS\system32> cd C:\Users\computer1\Desktop\Subnet-Calculator
PS C:\Users\computer1\Desktop\Subnet-Calculator>
```

Figure 1: Navigating to the Subnet-Calculator project directory using Windows PowerShell.

Step 4: Execute the PowerShell Script

The PowerShell script was executed using:

-*.\SubnetCalculator.ps1*

```
PS C:\Users\computer1\Desktop\Subnet-Calculator> .\SubnetCalculator.ps1
```

Figure 2: Executing the PowerShell script (SubnetCalculator.ps1).

Step 5: Resolve the Execution Policy Error

Initially, PowerShell displayed the following error:
Running scripts is disabled on this system.

To allow locally created PowerShell scripts to execute, Windows PowerShell was opened as Administrator and the following command was executed:

-*Set-ExecutionPolicy -Scope CurrentUser RemoteSigned*

The execution policy change was confirmed by typing:

-*Y*

The current execution policy was verified using:

-*Get-ExecutionPolicy*

Output:

RemoteSigned

Step 6: Execute the Script Again

After updating the execution policy, the PowerShell script was executed successfully using:

-*.\SubnetCalculator.ps1*

Step 7: Enter the Required Inputs

The application prompted for:
- IPv4 Address
- CIDR Prefix

Sample input used:

IPv4 Address : 192.168.206.126
CIDR Prefix : /24

Step 8: Perform Subnet Calculations

The program converted the CIDR prefix into the corresponding subnet mask and calculated:
- Subnet Mask
- Network Address
- Broadcast Address
- First Host Address
- Last Host Address
- Total Usable Hosts

Step 9: Verify the Results

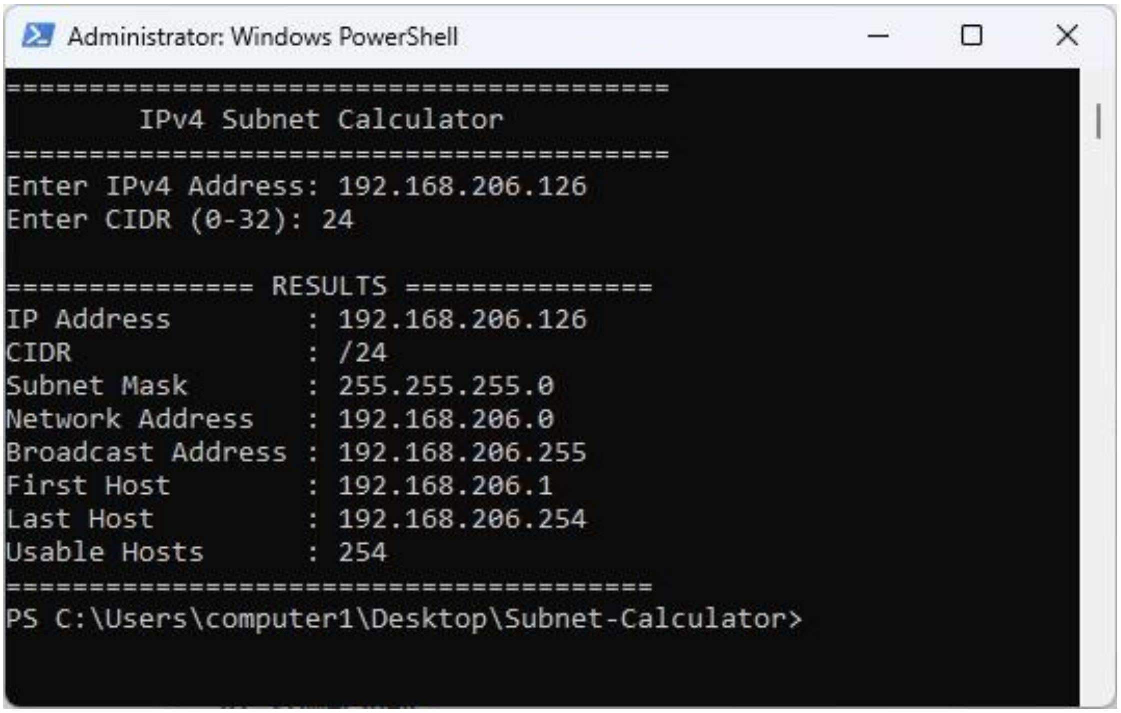
The generated subnetting information was verified using the sample input.

Input:
IPv4 Address : 192.168.206.126
CIDR : /24

Output:
Subnet Mask : 255.255.255.0
Network Address : 192.168.206.0
Broadcast Address : 192.168.206.255
First Host : 192.168.206.1
Last Host : 192.168.206.254
Usable Hosts : 254

The generated output matched the expected subnet calculations, confirming that the PowerShell subnet calculator was functioning correctly.

Execution Screenshot



Sample Output

Input:
IPv4 Address: 192.168.206.126
CIDR: /24

Output:
Subnet Mask: 255.255.255.0
Network Address: 192.168.206.0
Broadcast Address: 192.168.206.255
First Host: 192.168.206.1
Last Host: 192.168.206.254
Usable Hosts: 254

Conclusion

The experiment successfully demonstrated IPv4 subnet calculations using PowerShell. The application accurately generated subnet information and reinforced concepts such as CIDR notation, subnet masks, and address calculations. This experiment was completed as part of the CyberOps (Ethical Hacking) course assignment.